Practice Quiz: Lecture 8

Topics: Wireless Technologies, Spectrum Bands (ISM / UNII), Propagation Modes, Antenna Types, IEEE 802.11 Standards (g, ac, ax, ad, af), MIMO, and Wi-Fi Architecture

Student Name: _____ **Student ID:** _____ **Score:** _____ / 24

1. Among radio waves, microwaves, infrared, and light waves, which wireless transmission medium operates at the highest frequency range?

( ) A. Radio wave

( ) B. Microwave

( ) C. Infrared

( ) D. Light wave

2. Which IEEE 802.11 Wi-Fi standard operates in the 60 GHz millimeter-wave band and has the smallest communication range (typically limited to a single room)?

( ) A. IEEE 802.11b/g/n

( ) B. IEEE 802.11ah

( ) C. IEEE 802.11ad

( ) D. IEEE 802.11af

3. Which IEEE 802.11 standard ("White-Fi") operates in the unused television broadcast spectrum (VHF/UHF TV white spaces) to achieve the largest coverage area?

( ) A. IEEE 802.11ad

( ) B. IEEE 802.11af

( ) C. IEEE 802.11ac

( ) D. IEEE 802.11a

4. Which wireless technology is categorized as a Low-Power Wide Area Network (LPWAN) optimized for battery-operated IoT sensors?

Answer: _____

5. Which wireless technology (based on IEEE 802.16) is specifically classified as a Wireless Metropolitan Area Network (WMAN) technology?

Answer: _____

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6. Select all frequency bands that are unlicensed and standardized for commercial Wi-Fi communications:

[ ] A. 2.4 GHz (ISM band)

[ ] B. 3 GHz (VHF band)

[ ] C. 5 GHz (U-NII band)

[ ] D. 6 GHz (UHF band)

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7. In which operating mode does a Wi-Fi client device connect to network resources and the Internet through a centralized Access Point (AP)?

( ) A. Ad-hoc Mode (Peer-to-Peer)

( ) B. Infrastructure Mode

( ) C. Direct Mesh Mode

( ) D. Bridge Transponder Mode

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8. What is the defining characteristic of a "Stealth Access Point"?

( ) A. An AP that boosts wireless coverage through directional beamforming.

( ) B. An AP that cloaks its presence by disabling the public broadcasting of its SSID in beacon frames.

( ) C. An AP that requires no electrical power by scavenging RF energy.

( ) D. An AP that automatically re-encrypts all payloads using legacy WEP.

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9. Complete the frequency band and channel parameters:

An IEEE 802.11g access point operates in the _____ band and utilizes non-overlapping channels _____.

An IEEE 802.11ac access point operates in the _____ band and utilizes channels such as _____.

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10. What does Multiple-Input Multiple-Output (MIMO) technology accomplish in modern Wi-Fi standards (such as 802.11n, 802.11ac, and 802.11ax)?

( ) A. It forces devices to compress frames using public encryption keys.

( ) B. It utilizes multiple antennas to transmit and receive multiple parallel spatial data streams simultaneously.

( ) C. It restricts transmission exclusively to one single radio frequency channel.

( ) D. It shuts down unused radios to reduce electrical consumption during high traffic.

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11. Match each wireless application to its typical antenna radiation pattern:

Wireless Application	**Antenna Type (Omnidirectional / Unidirectional)**
1. Standard Indoor Wi-Fi Access Point	_____
2. IrDA (Infrared Data Association) link	_____
3. Point-to-Point Terrestrial Microwave Tower	_____

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12. Match each communication system with its primary electromagnetic propagation mode:

Communication System	**Propagation Mode (Ground / Sky / Line-of-Sight)**
1. Geostationary Satellite communication	_____
2. Commercial AM Radio broadcasting	_____
3. High-Frequency (HF) Amateur Ham Radio	_____
4. Commercial FM Radio broadcasting	_____
5. Handheld TV remote control	_____
6. 4G LTE and 5G Cellular networks	_____

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13. Why can commercial AM radio signals be received over vast geographic distances beyond the physical curvature of the Earth?

( ) A. AM signals propagate via low-frequency ground waves that hug the curvature of the Earth's surface.

( ) B. AM signals penetrate through the core of the Earth.

( ) C. AM signals use directional lasers that bounce off passing aircraft.

( ) D. AM signals utilize infrared line-of-sight relays.

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14. What atmospheric mechanism enables HF Ham Radio signals to travel thousands of kilometers intercontinentally via "sky propagation"?

( ) A. Gravitational lensing in the stratosphere.

( ) B. Refraction and reflection back to Earth by the charged ionosphere.

( ) C. Convection currents in the lower troposphere.

( ) D. Absorption by oceanic surface moisture.

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15. Why are consumer infrared (IrDA) remote control signals unable to control electronic appliances situated in an adjacent room?

( ) A. Infrared frequency is too low to oscillate through air.

( ) B. Infrared waves cannot penetrate solid, opaque barriers like walls and require strict line-of-sight.

( ) C. Infrared signals immediately curve downward due to ground propagation.

( ) D. Infrared transceivers require an active Wi-Fi handshake.

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16. In the 2.4 GHz ISM band, how many MHz wide is a standard 802.11b/g channel, and how far apart are channel center frequencies spaced?

Channel Bandwidth: _____ MHz

Channel Center Spacing: _____ MHz

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17. Which sub-1 GHz Wi-Fi protocol (operating in the 900 MHz band) was standardized specifically for long-range, low-power Smart Home and Internet-of-Things (IoT) deployments?

( ) A. IEEE 802.11ad

( ) B. IEEE 802.11ac

( ) C. IEEE 802.11ah (Wi-Fi HaLow)

( ) D. IEEE 802.11g

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18. According to the Free Space Path Loss (FSPL) formula, how does increasing transmission carrier frequency affect signal attenuation over a fixed distance?

( ) A. Higher frequencies experience greater path loss and attenuate more rapidly.

( ) B. Higher frequencies experience zero path loss in free space.

( ) C. Carrier frequency has no mathematical relationship to path attenuation.

( ) D. Lower frequencies attenuate significantly faster than higher frequencies.

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19. In Wi-Fi network terminology, what is the distinction between an SSID and a BSSID?

SSID: _____

BSSID: _____

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20. What multiplexing innovation introduced in IEEE 802.11ax (Wi-Fi 6) divides wireless channels into smaller Resource Units (RUs) to serve multiple concurrent clients?

( ) A. DSSS

( ) B. FHSS

( ) C. OFDMA (Orthogonal Frequency Division Multiple Access)

( ) D. CSMA/CD

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21. Satellite communications utilize microwave spectrum bands. Which propagation mode is strictly required between an Earth ground station dish and the orbiting satellite?

( ) A. Line-of-sight (LOS) propagation

( ) B. Sky wave ionospheric reflection

( ) C. Surface ground wave diffraction

( ) D. Tropospheric ducting

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22. What frequency classification encompass 4G LTE and 5G cellular communication signals?

( ) A. Exclusively audio sound waves

( ) B. Both Radio waves and Microwaves

( ) C. Exclusively visible light waves

( ) D. X-ray spectrum bands

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23. In an Ad-Hoc wireless network (Independent Basic Service Set - IBSS), which device is responsible for coordinating client-to-client communications?

( ) A. A dedicated root Access Point router.

( ) B. The client stations themselves communicate peer-to-peer without an intermediary AP.

( ) C. An upstream ISP DHCP server.

( ) D. An optical line terminal (OLT).

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24. What is the fundamental disadvantage of using 5 GHz Wi-Fi compared to 2.4 GHz Wi-Fi inside a residential home?

( ) A. 5 GHz has fewer non-overlapping channels than 2.4 GHz.

( ) B. 5 GHz signals have shorter wavelengths that attenuate more when passing through solid walls and doors, resulting in shorter physical range.

( ) C. 5 GHz supports significantly lower maximum bit rates than 2.4 GHz.

( ) D. 5 GHz experiences higher RF interference from domestic microwave ovens.

Answer Key & Detailed Explanations (Lecture 8)

Question 1: D [Light wave][cite: 13]

On the electromagnetic spectrum, frequency increases in the order: Radio waves (< 1 GHz) $\rightarrow$ Microwaves ($1 - 300$ GHz) $\rightarrow$ Infrared (300 GHz $- 400$ THz) $\rightarrow$ Light waves (> 400 THz)[cite: 13].

Question 2: C [IEEE 802.11ad][cite: 13]

IEEE 802.11ad (WiGig) utilizes the 60 GHz millimeter-wave band[cite: 13]. Because oxygen absorption and free-space path loss are severe at 60 GHz, its communication range is the smallest (typically 1 to 10 meters within line-of-sight in a single room)[cite: 13].

Question 3: B [IEEE 802.11af][cite: 14]

IEEE 802.11af operates in the unused television broadcast white spaces (sub-1 GHz, 54–790 MHz)[cite: 14]. Lower frequencies propagate much farther and penetrate walls efficiently, giving 802.11af the largest coverage area (up to several kilometers)[cite: 14].

Question 4: LoRa[cite: 14]

LoRa (Long Range) is a low-power, wide-area network (LPWAN) wireless modulation technology engineered for long-distance battery-powered IoT devices[cite: 14].

Question 5: WiMAX[cite: 14]

WiMAX (Worldwide Interoperability for Microwave Access, standardized under IEEE 802.16) is designed as a Wireless Metropolitan Area Network (WMAN) technology for broadband last-mile delivery[cite: 14].

Question 6: A and C [2.4 GHz (ISM), 5 GHz (UNII)][cite: 15]

The primary unlicensed frequency bands standardized for Wi-Fi are the 2.4 GHz Industrial, Scientific, and Medical (ISM) band and the 5 GHz Unlicensed National Information Infrastructure (U-NII) bands[cite: 15].

Question 7: B [Infrastructure Mode][cite: 15]

In Infrastructure Mode, wireless client stations communicate with the wired distribution network and Internet by associating through a central Access Point (BSS)[cite: 15].

Question 8: B [Hides presence by not broadcasting SSID][cite: 16]

A Stealth AP conceals its network name by suppressing the SSID field in periodic beacon frames; client devices must manually configure and probe for the SSID to connect[cite: 16].

Question 9: 2.4 GHz, channels [1, 6, 11]; 5 GHz, channels [42, 58, 106][cite: 16]

IEEE 802.11g operates at 2.4 GHz where the three standard non-overlapping channels are 1, 6, and 11[cite: 16]. IEEE 802.11ac operates at 5 GHz with wider 80 MHz channel numbers such as 42, 58, 106, and 122[cite: 16].

Question 10: B [Multiple antennas transmit and receive multiple streams simultaneously][cite: 17]

MIMO (Multiple-Input Multiple-Output) exploits multipath propagation by employing multiple antennas to transmit and decode distinct spatial data streams at the same time, multiplying link capacity[cite: 17].

Question 11: Matching Antennas[cite: 17]

1. Wi-Fi AP $\rightarrow$ **Omnidirectional antenna** (360-degree coverage)[cite: 17]
2. IrDA $\rightarrow$ **Unidirectional antenna / focused beam**[cite: 17]
3. Microwave links $\rightarrow$ **Unidirectional antenna** (parabolic dish)[cite: 17]

Question 12: Propagation Modes[cite: 18]

1. Satellite $\rightarrow$ **Line-of-Sight propagation** (Microwave)[cite: 18]
2. AM Radio $\rightarrow$ **Ground propagation**[cite: 18]
3. Ham Radio $\rightarrow$ **Sky propagation** (Ionospheric bounce)[cite: 18]
4. FM Radio $\rightarrow$ **Line-of-Sight propagation**[cite: 18]
5. TV Remote $\rightarrow$ **Line-of-Sight propagation** (Infrared)[cite: 18]
6. 4G/5G Cellular $\rightarrow$ **Line-of-Sight propagation** (Radio/Microwave)[cite: 18]

Question 13: A[cite: 18]

AM broadcast frequencies (530 to 1700 kHz) travel via ground wave propagation, where low-frequency electromagnetic waves follow the curvature of the Earth along the conducting surface[cite: 18].

Question 14: B[cite: 18]

In sky propagation, High-Frequency (HF) waves (3–30 MHz) radiate upward and are bent/reflected back toward the ground by ionized gas layers in the ionosphere, allowing transcontinental skip[cite: 18].

Question 15: B[cite: 18]

Infrared light has very short wavelengths that are absorbed or reflected by opaque objects; it cannot penetrate structural drywall or concrete, confining signals strictly to line-of-sight in the room[cite: 18].

Question 16: $20\ \mathrm{MHz}$ (or $22\ \mathrm{MHz}$) Bandwidth; $5\ \mathrm{MHz}$ Spacing

In the 2.4 GHz band, channels are allocated center frequencies spaced $5\ \mathrm{MHz}$ apart, but individual OFDM/DSSS channels occupy $20\ \mathrm{MHz}$ to $22\ \mathrm{MHz}$ of spectral width, causing adjacent channels to overlap.

Question 17: C [IEEE 802.11ah]

IEEE 802.11ah (commercialized as Wi-Fi HaLow) operates in the sub-gigahertz 900 MHz band to provide long range (up to 1 km) and low energy usage for massive IoT networks.

Question 18: A [Higher frequencies experience greater path loss]

Free-space path loss formula:

$$\mathrm{FSPL} = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right)$$

As carrier frequency f increases, FSPL increases by $20\log_{10}(f)$, causing higher frequencies to attenuate significantly faster over distance.

Question 19: SSID = Network Name; BSSID = Radio MAC Address

The Service Set Identifier (SSID) is the human-readable text name of the wireless network. The Basic Service Set Identifier (BSSID) is the hardware 48-bit MAC address of the Access Point radio interface.

Question 20: C [OFDMA]

Wi-Fi 6 (802.11ax) introduced OFDMA, which subdivides an RF channel into orthogonal subcarriers grouped into Resource Units (RUs), enabling the AP to transmit to or receive from multiple clients simultaneously.

Question 21: A [Line-of-sight propagation][cite: 18]

Microwave signals above 1 GHz pass through the ionosphere without reflecting, requiring a direct, unblocked Line-of-Sight between the ground dish and satellite transponder[cite: 18].

Question 22: B [Both Radio waves and Microwaves][cite: 18]

Cellular standards utilize lower UHF radio bands (600 MHz – 1 GHz) as well as microwave/millimeter-wave bands (1.8 GHz – 39 GHz)[cite: 18].

Question 23: B [Client stations communicate peer-to-peer without an AP]

In an Ad-Hoc (IBSS) wireless network topology, peer devices connect directly to each other without utilizing a centralized Access Point.

Question 24: B

5 GHz signals suffer greater attenuation through walls and building materials than 2.4 GHz signals due to their shorter wavelength and higher path loss, yielding smaller indoor coverage.
